



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: SOA PROGRAMMING AND MICROSERVICES– (24SDCS03)

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TITLE OF THE PROJECT	Task & Project Management: Project Deadline Reminder System	
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Abstract:

The Task & Project Management: AI-Powered Project Deadline Reminder System is a web-based application designed to help individuals and teams efficiently manage projects, tasks, priorities, and deadlines. Managing multiple tasks manually can lead to missed deadlines, poor prioritization, delayed project completion, and reduced productivity.

The proposed system provides a centralized platform where users can create projects, add tasks, assign priorities, set deadlines, track task progress, and receive automated reminders for upcoming and overdue tasks. In addition to conventional task management, the system integrates Artificial Intelligence (AI) to intelligently analyze tasks based on factors such as deadline, priority, estimated effort, dependencies, and current progress. The AI recommends which tasks should be completed first and provides explanations for its recommendations.

The system also uses an automated scheduling mechanism to continuously monitor deadlines and generate timely reminders. The application follows a three-tier architecture, consisting of a React.js frontend, Spring Boot backend, and MySQL database, with an AI service integrated into the

backend. REST APIs are used for communication between the frontend, backend, database, and AI service.

The main objective of the project is to combine traditional project management, automated deadline reminders, and AI-based task prioritization into a single platform that helps users make better decisions, reduce missed deadlines, and improve overall productivity.

Problem Statement:

Managing multiple projects and tasks with different priorities, deadlines, and workloads can become difficult when users rely on manual methods such as notebooks, spreadsheets, or basic task lists.

Users may know the deadlines of their tasks but may not know which task should be completed first. A task with a later deadline may require more effort or may block several other tasks. Similarly, users may fail to identify projects that are at risk of missing their deadlines.

Existing basic task management systems generally provide features such as task creation, status tracking, and deadline notifications, but they often do not provide intelligent recommendations based on the user's workload and task characteristics.

Therefore, there is a need for an intelligent project management system that can:

- Centralize project and task management.
- Track task deadlines and progress.
- Automatically remind users about upcoming and overdue deadlines.
- Analyze task characteristics and recommend task priorities.
- Help users identify high-risk or urgent tasks.
- Provide meaningful recommendations to improve project completion.

Proposed Solution:

The proposed system provides a centralized web-based platform for managing projects and tasks while using AI to assist users in making better task-prioritization decisions.

A. Project Management

Users can:

- Create projects.
- Edit and delete projects.
- Set project deadlines.
- Track project completion.
- View active and completed projects.

B. Task Management

Users can:

- Create tasks under projects.
- Set deadlines.
- Assign priority.
- Specify estimated effort.
- Track task status.
- Mark tasks as completed.
- Manage pending and overdue tasks.

C. AI-Based Task Prioritization

This is the main AI component of the project.

The system sends task information to the AI engine, including:

- Task description.
- Deadline.
- Existing priority.
- Estimated effort.
- Dependencies.
- Current project progress.

The AI analyzes these factors and recommends the order in which tasks should be completed.

AI Recommendation: Complete this task first because it has a close deadline, high effort, and may affect other project activities.

The system can display the recommendation directly on the dashboard.

D. Automated Deadline Reminder

A scheduler periodically checks task deadlines.

The system can support in-app notifications and can later be extended to email notifications.

E. AI Productivity Recommendations

The AI can also provide recommendations based on the user's pending workload.

